

Dynamic Ticket Pricing at FedEx Forum

Evidence from the 2024–25 Memphis Grizzlies Season

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Introduction: Why NBA Ticket Pricing?

Problem: Pricing a Differentiated, Perishable Product

Dynamic Pricing in Sports:

- **Definition:** Demand-based pricing strategy
- Widely used in industries such as airlines, hospitality, and sports
- Objective: Align prices with consumer willingness to pay (Kopalle et al., 2023)
- Common demand signals include:
 - Opponent quality and team performance
 - Injuries to star players
 - Time remaining until game

This Paper: Four research questions across 729,554 seat–game observations at FedEx Forum

Data: FedEx Forum, 2024–25 Season

Dataset Overview:

- **729,554 observations** — every seat $\times$ every game
- **41 home games**, October 27, 2024 – March 24, 2025
- **17,794 seats** across 79 sections
- Prices observed at **three time points**: season-start, two weeks out, tipoff

Seating Levels:

Level	Seats	Median Tipoff Price
Courtside	246	\$1,460
Lower Bowl	7,843	\$182
Club	2,050	\$256
Upper Bowl	7,655	\$51

Key game-level variables: opponent win %, Vegas O/U, high-demand flag, injury indicators

Data: Variable Overview

#	Name	Varies by
1	section	seat
2	row	seat
3	seat	seat
4	level	seat
5	game_id	game
6	event_date	game
7	day_of_week	game
8	month	game
9	opponent	game
10	opp_preseason_ou	game
11	opp_win_pct_2wk	game
12	high_demand	game
13	is_rivalry	game
14	opp_star_injured	game
15	grizz_star_injured	game
16	price_season_start	seat $\times$ game
17	price_2wk_before	seat $\times$ game
18	price_tipoff	seat $\times$ game
19	attendance	game

Sample rows (Section 101, Game 1):

Row	Seat	P _{start}	P _{2wk}	P _{tipoff}
A	1	\$155	\$185	\$212
A	2	\$144	\$161	\$164
A	3	\$158	\$182	\$181
A	4	\$140	\$172	\$174
A	5	\$150	\$175	\$192
A	6	\$144	\$191	\$197
A	7	\$139	\$163	\$164
A	8	\$163	\$155	\$156
A	9	\$166	\$214	\$211
A	10	\$144	\$172	\$173

- Game 1: vs. Orlando Magic, Oct 27 2024
- **Opponent:** O/U = 38.5, Win% = 0.512
- All logical flags FALSE for this game
- Attendance: 14,527

Methods: Log-Linear Dynamic Pricing Model

Core idea: Log-differences cancel seat fixed effects:

$$\Delta_i^{2wk} = \log(P_i^{2wk}) - \log(P_i^{start})$$

Four estimation steps:

- ➊ **Step 1 — Season-Start Price:** Regress $\log(P^{start})$ on seat level + game controls. Standard errors clustered by game.
- ➋ **Step 2 — Start → Two Weeks Out:** Regress Δ^{2wk} on game-level demand shifters (N=41, HC3 robust SEs).
- ➌ **Step 3 — Two Weeks Out → Tipoff:** Regress Δ^{tip} on game-level variables + seat-level interactions.
- ➍ **Step 4 — Attendance:** Linear model on 41-game dataset with robust SEs.

Software: R (`fixest`, `lmtest`, `sandwich`, `modelsummary`)

Findings: Key Results Across All Four Steps

Step 1 — Season-Start

- Seat level explains **95%** of price variation ($R^2 = 0.95$)
- Courtside: **+2,733%** vs Upper Bowl
- Each opponent win $\Rightarrow$ **+0.18%** price

Step 3 — To Tipoff

- Effects **smaller** — market already adjusted
- Rivalry premium **fully absorbed** by 2 weeks out
- Lower Bowl: extra **+8%** on high demand games

Step 2 — To Two Weeks Out

- High demand: **+22%** price increase
- Rival games: **+232%** appreciation
- Opp. star injured: **-18%**
- Grizz star injured: **-78%**

Step 4 — Attendance

- High demand: **+2,405 fans**
- Opp. star injured: **-1,484 fans**
- Grizz star injured: **-794 fans**
- Saturday/Sunday: **+830-898 fans**

Results: What Drives Season-Start Prices?

Table 1: Step 1 — What Drives Season-Start Prices?

	(1A) Game FE Log Price Start	(1B) Full Log Price Start
Lower Bowl	1.260*** (0.000)	1.260*** (0.000)
Club	1.602*** (0.000)	1.602*** (0.000)
Courtside	3.345*** (0.000)	3.345*** (0.000)
Intercept		3.728*** (0.000)
opp_preseason_ou		0.002*** (0.000)
high_demand		0.000 (0.000)
is_rivalry		−0.000 (0.000)
Day of Week FE	No	Yes
Month FE	No	Yes
Game FE	Yes	No
Observations	729,554	729,554
R ²	0.950	0.950
Adj. R ²	0.950	0.950
RMSE	0.17	0.17

Notes: Standard errors clustered by `game_id` in parentheses. Upper Bowl is omitted category. † $p < 0.1$,
 * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

What Did We Learn? Why Does This Matter?

Key Takeaways:

- **Seat quality dominates** season-start prices (hedonic pricing)
- **Market is efficient** — most info priced in by two weeks out
- **Star injuries suppress prices** — Grizzlies -78% vs opponent -18%
- **Premium seats react differently** to marquee games

Why It Matters:

- Helps teams **optimize revenue** via smarter dynamic pricing
- Quantifies the **economic value** of star players
- Shows how secondary markets **efficiently incorporate information**

Future Work: multiple seasons, primary market data, IV estimation

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